

Qualification Pack



Blockchain Architect

QP Code: SSC/Q8702

Version: 3.0

NSQF Level: 6

IT-ITeS SSC NASSCOM || NASSCOM Plot No - 7, 8, 9 & 10, 3rd Floor, Sector 126
Noida Uttar Pradesh - 201303 || email:sscstandards@nasscom.in

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SSC/Q8702: Blockchain Architect

Brief Job Description

The Blockchain Architect is responsible for designing, developing, and maintaining blockchain-based systems. This role involves defining project requirements, designing system architecture, and developing smart contracts focusing on security. The developer must assess vulnerabilities, ensure scalability, and integrate blockchain solutions with existing systems. Additionally, they establish governance models and adhere to best practices in blockchain management.

Personal Attributes

Individuals in this job role must have coordination skills to interact with multiple stakeholders as well as team members and to ensure project success. They must possess good communication skills to articulate their arguments and persuade others. They should have a professional commitment to constantly update their knowledge on the latest technological updates.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N8732: Define project requirements, conduct feasibility studies, and select appropriate blockchain platform.](#)
2. [SSC/N8733: Design overall system architecture, including data flow, consensus mechanism, and node structure.](#)
3. [SSC/N8734: Develop high-level smart contract specifications and security strategies & deploy the blockchain solution](#)
4. [SSC/N8735: Identify vulnerabilities in various layers of blockchain and develop security features for them.](#)
5. [SSC/N8736: Plan for scalability and integration with existing systems](#)
6. [SSC/N8737: Establish governance models and best practices](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Future Skills

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Occupation	Blockchain
Country	India
NSQF Level	6
Credits	19
Aligned to NCO/ISCO/ISIC Code	NCO-2015/NIL
Minimum Educational Qualification & Experience	Completed 4 year UG program (PG/UG/Degree/Diploma in (Engg./Science)) OR Completed 3 year UG degree (PG/UG/Degree/Diploma in (Engg./Science)) with 1.5 years of experience in relevant field. Relevant experience in job roles related to Blockchain/Computer Science/IT The relevant experience would include work, internship and apprenticeship after completing relevant educational qualifications. OR Previous relevant Qualification of NSQF Level (5) with 1.5 years of experience in relevant field. Relevant experience in job roles related to Blockchain/Computer Science/IT The relevant experience would include work, internship and apprenticeship after completing relevant educational qualifications.
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	08/05/2028
NSQC Approval Date	08/05/2025
Version	3.0
Reference code on NQR	QG-06-IT-01844-2025-V2-NASSCOM
NQR Version	3.0

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SSC/N8732: Define project requirements, conduct feasibility studies, and select appropriate blockchain platform.

Description

This unit is about evaluating the feasibility of blockchain adoption and formulating an initial roadmap

Scope

The scope covers the following :

- Define Project Requirements
- Conduct Feasibility Study
- Select Appropriate Blockchain Platform
- Create Implementation Roadmap

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Identify the specific business problem or use case (e.g. product tracability,digital identity,NFT,insurance claim processing etc.) you want to address with blockchain
- PC2.** Outline functional and non-functional requirements for the solution
- PC3.** Define key objectives, success criteria, and expected outcomes
- PC4.** Determine privacy, security, scalability and performance needs
- PC5.** Identify integration requirements with existing systems
- PC6.** Define regulatory compliance needs
- PC7.** Analyze if blockchain is truly needed or if traditional databases would suffice
- PC8.** Evaluate technical feasibility - can blockchain meet performance/scalability needs?
- PC9.** Assess economic feasibility - costs vs benefits, ROI analysis
- PC10.** Examine operational feasibility - impact on existing processes/systems
- PC11.** Evaluate legal/regulatory feasibility - compliance with relevant regulations
- PC12.** Analyze organizational readiness - skills, culture, change management needs
- PC13.** Determine public vs private blockchain needs
- PC14.** Evaluate consensus mechanisms (PoW, PoS, etc.)
- PC15.** Assess programming languages and development tools
- PC16.** Compare transaction speeds and costs
- PC17.** Evaluate smart contract capabilities
- PC18.** Assess interoperability with other systems
- PC19.** Compare security features and track record
- PC20.** Evaluate community support and ecosystem
- PC21.** Consider adoption rates and long-term viability
- PC22.** Define project phases and milestones
- PC23.** Allocate resources and budget

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PC24. Create risk mitigation plans

PC25. Define governance model

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Proficiency in multiple programming languages. Especially, Solidity (for Ethereum smart contracts), Python, Java, JavaScript, C plus plus etc.
- KU2.** Web development expertise (front-end and back-end)
- KU3.** Experience with blockchain-specific languages like Golang
- KU4.** Deep understanding of blockchain platforms and blockchain architecture and concept like DLT, Consensus mechanisms (e.g., Proof of Work, Proof of Stake), Public, private, and consortium blockchains, Smart contracts, Dapps
- KU5.** Data Structures and Algorithms
- KU6.** Understanding of cryptographic concepts like Public key cryptography, Digital signatures, Symmetric/asymmetric encryption, Cryptographic hash functions (e.g., SHA-256) etc.
- KU7.** Knowledge of interoperability between different blockchain networks
- KU8.** Understanding of blockchain security best practices
- KU9.** Knowledge of Project Management and Analysis
- KU10.** Understanding of relevant industry use cases for blockchain
- KU11.** Ability to translate business requirements into technical specifications
- KU12.** Knowledge of relevant regulations (e.g., AML/CFT, KYC, GDPR, HIPAA)
- KU13.** Knowledge of DevOps practices and tools, API Development, Cloud platforms (e.g., AWS Blockchain), Version control system (Github etc.)

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Articulate complex blockchain concepts to non-technical stakeholders
- GS2.** Foster teamwork between technical and business teams
- GS3.** Break down complex blockchain-related issues into manageable parts
- GS4.** Flexibility in adjusting roadmaps as new developments emerge
- GS5.** Finding common ground between different departments' requirements

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Identify the specific business problem or use case (e.g. product tracability,digital identity,NFT,insurance claim processing etc.) you want to address with blockchain	1	2	-	0.5
PC2. Outline functional and non-functional requirements for the solution	1	2	-	0.5
PC3. Define key objectives, success criteria, and expected outcomes	1	2	-	0.5
PC4. Determine privacy, security, scalability and performance needs	1	2	-	0.5
PC5. Identify integration requirements with existing systems	1	2	-	0.5
PC6. Define regulatory compliance needs	1	2	-	0.5
PC7. Analyze if blockchain is truly needed or if traditional databases would suffice	1	2	-	0.5
PC8. Evaluate technical feasibility - can blockchain meet performance/scalability needs?	1	2	-	0.5
PC9. Assess economic feasibility - costs vs benefits, ROI analysis	1	2	-	0.5
PC10. Examine operational feasibility - impact on existing processes/systems	1	2	-	0.5
PC11. Evaluate legal/regulatory feasibility - compliance with relevant regulations	1	2	-	1
PC12. Analyze organizational readiness - skills, culture, change management needs	1	2	-	1
PC13. Determine public vs private blockchain needs	1	2	-	1
PC14. Evaluate consensus mechanisms (PoW, PoS, etc.)	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. Assess programming languages and development tools	1	2	-	1
PC16. Compare transaction speeds and costs	1	2	-	1
PC17. Evaluate smart contract capabilities	1	2	-	1
PC18. Assess interoperability with other systems	1	2	-	1
PC19. Compare security features and track record	1	2	-	1
PC20. Evaluate community support and ecosystem	1	2	-	1
PC21. Consider adoption rates and long-term viability	2	2	-	1
PC22. Define project phases and milestones	2	2	-	1
PC23. Allocate resources and budget	2	2	-	1
PC24. Create risk mitigation plans	2	2	-	1
PC25. Define governance model	2	2	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8732
NOS Name	Define project requirements, conduct feasibility studies, and select appropriate blockchain platform.
Sector	IT-ITeS
Sub-Sector	
Occupation	Blockchain
NSQF Level	6
Credits	1
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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SSC/N8733: Design overall system architecture, including data flow, consensus mechanism, and node structure.

Description

This NOS is about developing the overall system architecture for the blockchain solution

Scope

The scope covers the following :

- System Architecture Design
- Network layer
- Consensus layer
- Data layer
- Execution layer
- Application Layer

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Design the overall system architecture, including on-chain and off-chain components
- PC2.** Design the network architecture that supports the blockchain application, including node configuration, network protocols, and communication channels
- PC3.** Evaluate different consensus mechanisms like Proof of Work (PoW), Proof of Stake (PoS), or Delegated Proof of Stake (DPoS) based on security, decentralization, scalability, and energy efficiency
- PC4.** Define the structure and flow of blockchain data within the blockchain network
- PC5.** Define the execution layer for processing of transactions and smart contracts.
- PC6.** Define the application layer, it provides interfaces for users and applications to interact with the blockchain.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Deep understanding of blockchain fundamentals, including consensus mechanisms, cryptography, and distributed ledger technology
- KU2.** Proficiency in creating high-level and low-level system designs
- KU3.** Understanding of distributed computing principles
- KU4.** Knowledge of networking protocols and standards
- KU5.** Proficiency in designing both on-chain and off-chain data storage solutions
- KU6.** Expertise in cryptographic protocols and their implementation
- KU7.** Proficiency in designing RESTful APIs and GraphQL interfaces
- KU8.** Ability to design high-performance blockchain systems

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- KU9.** Understanding of cross-chain communication protocols
- KU10.** Proficiency in designing cloud-native blockchain solutions
- KU11.** Understanding of DevOps principles and practices
- KU12.** Proficiency in languages commonly used in blockchain development (e.g., Solidity, Go, Rust)
- KU13.** Understanding of Layer 2 scaling solutions (e.g., state channels, sidechains)
- KU14.** Ability to design on-chain governance mechanisms

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to clearly articulate complex technical concepts to both technical and non-technical stakeholders.
- GS2.** Skill to analyze problems objectively and make logical decisions based on available information.
- GS3.** Capacity to think outside the box and develop innovative architectural solutions.
- GS4.** Flexibility to adjust designs and approaches as new blockchain technologies and requirements emerge.
- GS5.** Aptitude for working effectively with cross-functional teams to integrate diverse perspectives into the architecture.
- GS6.** Capability to guide and inspire teams throughout the architecture development process.
- GS7.** Proficiency in identifying and resolving complex architectural challenges efficiently.
- GS8.** Meticulousness in considering all aspects of the system design, from high-level concepts to minute technical details.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Design the overall system architecture, including on-chain and off-chain components	6	9	-	4
PC2. Design the network architecture that supports the blockchain application, including node configuration, network protocols, and communication channels	5	9	-	4
PC3. Evaluate different consensus mechanisms like Proof of Work (PoW), Proof of Stake (PoS), or Delegated Proof of Stake (DPoS) based on security, decentralization, scalability, and energy efficiency	6	8	-	3
PC4. Define the structure and flow of blockchain data within the blockchain network	5	8	-	3
PC5. Define the execution layer for processing of transactions and smart contracts.	4	8	-	3
PC6. Define the application layer, it provides interfaces for users and applications to interact with the blockchain.	4	8	-	3
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8733
NOS Name	Design overall system architecture, including data flow, consensus mechanism, and node structure.
Sector	IT-ITeS
Sub-Sector	
Occupation	Blockchain
NSQF Level	6
Credits	5
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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SSC/N8734: Develop high-level smart contract specifications and security strategies & deploy the blockchain solution

Description

This NOS is about high-level smart contract specifications and security strategies

Scope

The scope covers the following :

- Determine Contract Architecture
- Implement Security Strategies

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Define Contract Purpose and Functionality (For eg.Token creation and management,Decentralized exchange mechanisms,Supply chain tracking,Decentralized finance (DeFi) protocols)
- PC2.** Identify key components and their interactions
- PC3.** Determine inheritance patterns and contract relationships
- PC4.** Define the core data that the contract will manage
- PC5.** Specify roles and permissions within contract
- PC6.** Implement Secure Coding Practices
- PC7.** Write extensive unit tests for all functions
- PC8.** Conduct multiple rounds of audits
- PC9.** Implement Fail-Safe Mechanisms
- PC10.** Ensure the security of any off-chain elements interacting with the contract
- PC11.** Implement real-time monitoring for suspicious activities
- PC12.** Use multi-signature wallets for critical functions and deploy the solution

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Deep understanding of blockchain fundamentals, including consensus mechanisms, cryptography, and distributed ledger technology
- KU2.** Proficiency in smart contract programming and deployment
- KU3.** Experience with development environments like Remix, Truffle, Hardhat, or Brownie
- KU4.** Familiarity with Web3.js, ethers.js, or other blockchain interaction libraries
- KU5.** Understanding of version control system
- KU6.** Knowledge of secure coding practices and vulnerability assessment
- KU7.** Experience with smart contract auditing and risk assessment

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- KU8.** Proficiency in testing frameworks and debugging tools specific to blockchain development
- KU9.** Knowledge of blockchain-specific security measures and best practices
- KU10.** Understanding of compliance requirements and security controls for blockchain solutions

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to clearly articulate complex technical concepts to both technical and non-technical stakeholders.
- GS2.** Skill to analyze problems objectively and make logical decisions based on available information.
- GS3.** Capacity to think outside the box and develop innovative architectural solutions.
- GS4.** Flexibility to adjust designs and approaches as new blockchain technologies and requirements emerge.
- GS5.** Aptitude for working effectively with cross-functional teams to integrate diverse perspectives into the architecture.
- GS6.** Proficiency in identifying and resolving complex architectural challenges efficiently.
- GS7.** Meticulousness in considering all aspects of the system design, from high-level concepts to minute technical details.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Define Contract Purpose and Functionality (For eg.Token creation and management,Decentralized exchange mechanisms,Supply chain tracking,Decentralized finance (DeFi) protocols)	3	6	-	1
PC2. Identify key components and their interactions	3	6	-	1
PC3. Determine inheritance patterns and contract relationships	3	5	-	1
PC4. Define the core data that the contract will manage	3	5	-	2
PC5. Specify roles and permissions within contract	3	5	-	1
PC6. Implement Secure Coding Practices	3	5	-	2
PC7. Write extensive unit tests for all functions	3	4	-	2
PC8. Conduct multiple rounds of audits	3	4	-	2
PC9. Implement Fail-Safe Mechanisms	3	4	-	2
PC10. Ensure the security of any off-chain elements interacting with the contract	1	2	-	2
PC11. Implement real-time monitoring for suspicious activities	1	2	-	2
PC12. Use multi-signature wallets for critical functions and deploy the solution	1	2	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8734
NOS Name	Develop high-level smart contract specifications and security strategies & deploy the blockchain solution
Sector	IT-ITeS
Sub-Sector	
Occupation	Blockchain
NSQF Level	6
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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SSC/N8735: Identify vulnerabilities in various layers of blockchain and develop security features for them.

Description

This unit is about securing the blockchain solution

Scope

The scope covers the following :

- Secure the core blockchain
- Create relevant documents and reports

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Collaborate with the core blockchain development team to secure various layers of blockchain (data layer, network layer, consensus layer, incentive layer, smart contract layer, etc.) and incorporate security features in them
- PC2.** Develop security monitoring and management controls for core blockchain
- PC3.** Conduct security and vulnerability assessments, including penetration testing, to identify risks and weaknesses in core blockchain
- PC4.** Perform security testing to ensure that proper cryptographic encryption and consensus mechanism is followed for each transaction through individual nodes
- PC5.** Research on latest blockchain attacks as well as the methods to prevent them
- PC6.** Support and guide incident response activities
- PC7.** Act as a subject matter expert for queries related to blockchain security
- PC8.** Provide strategic guidance by developing data-backed reports on security risks, recommendations, standard operating procedures, policies, controls and programs pertaining to blockchain security
- PC9.** Create documents and develop a knowledge base of threat models, security risks and their mitigation plans
- PC10.** Prepare reports on existing security posture and suitable recommendations for various stakeholders

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Deep understanding of blockchain fundamentals, including consensus mechanisms, cryptography, and distributed ledger technology
- KU2.** Proficiency in creating high-level and low-level system designs
- KU3.** Understanding of distributed computing principles
- KU4.** Knowledge of networking protocols and standards

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- KU5.** Proficiency in designing both on-chain and off-chain data storage solutions
- KU6.** Expertise in cryptographic protocols and their implementation
- KU7.** Proficiency in designing RESTful APIs and GraphQL interfaces
- KU8.** Ability to design high-performance blockchain systems
- KU9.** Understanding of cross-chain communication protocols
- KU10.** Proficiency in designing cloud-native blockchain solutions
- KU11.** Understanding of DevOps principles and practices
- KU12.** Proficiency in languages commonly used in blockchain development (e.g., Solidity, Go, Rust)
- KU13.** Understanding of Layer 2 scaling solutions (e.g., state channels, sidechains)
- KU14.** Ability to design on-chain governance mechanisms

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to clearly articulate complex technical concepts to both technical and non-technical stakeholders.
- GS2.** Skill to analyze problems objectively and make logical decisions based on available information.
- GS3.** Capacity to think outside the box and develop innovative architectural solutions.
- GS4.** Flexibility to adjust designs and approaches as new blockchain technologies and requirements emerge.
- GS5.** Aptitude for working effectively with cross-functional teams to integrate diverse perspectives into the architecture.
- GS6.** Capability to guide and inspire teams throughout the architecture development process.
- GS7.** Proficiency in identifying and resolving complex architectural challenges efficiently.
- GS8.** Meticulousness in considering all aspects of the system design, from high-level concepts to minute technical details.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Collaborate with the core blockchain development team to secure various layers of blockchain (data layer, network layer, consensus layer, incentive layer, smart contract layer, etc.) and incorporate security features in them	5	7	-	2
PC2. Develop security monitoring and management controls for core blockchain	3	5	-	2
PC3. Conduct security and vulnerability assessments, including penetration testing, to identify risks and weaknesses in core blockchain	4	6	-	2
PC4. Perform security testing to ensure that proper cryptographic encryption and consensus mechanism is followed for each transaction through individual nodes	4	6	-	2
PC5. Research on latest blockchain attacks as well as the methods to prevent them	3	5	-	2
PC6. Support and guide incident response activities	2	4	-	2
PC7. Act as a subject matter expert for queries related to blockchain security	2	4	-	2
PC8. Provide strategic guidance by developing data-backed reports on security risks, recommendations, standard operating procedures, policies, controls and programs pertaining to blockchain security	3	5	-	2
PC9. Create documents and develop a knowledge base of threat models, security risks and their mitigation plans	2	4	-	2
PC10. Prepare reports on existing security posture and suitable recommendations for various stakeholders	2	4	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8735
NOS Name	Identify vulnerabilities in various layers of blockchain and develop security features for them.
Sector	IT-ITeS
Sub-Sector	
Occupation	Blockchain
NSQF Level	6
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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SSC/N8736: Plan for scalability and integration with existing systems

Description

The NOS is about the integration and scalability

Scope

The scope covers the following :

- Integration with existing systems
- Scalability Enhancement

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Implement blockchain integration gradually to minimize disruption to existing system
- PC2.** Create middleware to facilitate communication between the blockchain and legacy systems
- PC3.** Leverage APIs for seamless integration between blockchain and existing systems
- PC4.** Implement industry-standard protocols for data exchange (Eg. Interoperability solutions like Polkadot or Cosmos for cross-chain communication)
- PC5.** Develop adapters for legacy systems that cannot directly interact with the blockchain
- PC6.** Implement Layer 2 scaling solutions to offload transaction processing from the main blockchain (for eg. state chain, sidechains, Plasma chains)
- PC7.** Implement sharding to improve scalability by partitioning the blockchain network

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding of blockchain fundamentals (distributed ledger technology Smart contracts etc.)
- KU2.** Expertise in legacy Programming Languages, databases, and mainframe systems
- KU3.** Skills in creating and managing APIs, including RESTful APIs and GraphQL.
- KU4.** Knowledge of enterprise service buses (ESBs) and other middleware platforms
- KU5.** Skills in handling and analyzing data across systems
- KU6.** Knowledge of Data protection laws like GDPR or CCPA and Industry specific regulations (eg HIPAA for healthcare, PCI DSS for financial service)

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to tackle complex technical challenges in blockchain systems.
- GS2.** Meticulousness in identifying and addressing potential issues.

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- GS3.** Flexibility to keep up with rapidly evolving blockchain technologies.
- GS4.** Skill to explain technical concepts to both technical and non-technical stakeholders.
- GS5.** Ability to work effectively with diverse teams, including developers and business analysts.
- GS6.** Capacity to analyze and evaluate different testing approaches and results.
- GS7.** Persistence in debugging and troubleshooting complex blockchain interactions.
- GS8.** Efficiency in prioritizing and executing various testing tasks.
- GS9.** Enthusiasm for staying updated with the latest blockchain testing methodologies.
- GS10.** Innovative thinking in designing test scenarios for unique blockchain use cases.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Implement blockchain integration gradually to minimize disruption to existing system	5	7	-	3
PC2. Create middleware to facilitate communication between the blockchain and legacy systems	5	7	-	3
PC3. Leverage APIs for seamless integration between blockchain and existing systems	4	7	-	3
PC4. Implement industry-standard protocols for data exchange (Eg. Interoperability solutions like Polkadot or Cosmos for cross-chain communication)	5	7	-	3
PC5. Develop adapters for legacy systems that cannot directly interact with the blockchain	4	7	-	3
PC6. Implement Layer 2 scaling solutions to offload transaction processing from the main blockchain (for eg. state chain, sidechains, Plasma chains)	4	7	-	3
PC7. Implement sharding to improve scalability by partitioning the blockchain network	3	8	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8736
NOS Name	Plan for scalability and integration with existing systems
Sector	IT-ITeS
Sub-Sector	
Occupation	Blockchain
NSQF Level	6
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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SSC/N8737: Establish governance models and best practices

Description

This unit is about aligning the blockchain solution to establish governance models and best practices

Scope

The scope covers the following :

- Evaluate external (regulatory and governmental) standards
- Evaluate internal (organizational) standards
- Drive blockchain governance
- Drive thought leadership initiatives

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Incorporate regulatory standards that are applicable for the blockchain solution based on its use case (HIPAA for healthcare industry, etc) in the solution design
- PC2.** Ensure compliance with the government regulations, and plan audits/ reviews to monitor the same
- PC3.** Assess the policies, standards and guidelines of the organization applicable to blockchain implementation
- PC4.** Define enterprise processes to ensure satisfactory performance of the solution
- PC5.** Incorporate on-chain and off-chain governance standards
- PC6.** Define a plan for conducting code audits and security compliance, and define standards for the same
- PC7.** Establish governance models for decentralized systems.
- PC8.** Ensure compliance with relevant regulations (e.g., GDPR, KYC/AML)
- PC9.** Implement necessary measures for data privacy and protection.
- PC10.** Conduct research to identify the latest developments in blockchain technologies by working with the R&D team
- PC11.** Identify opportunities for blockchain implementation and its adoption to new use cases
- PC12.** Create whitepapers collating the research data

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Deep understanding of blockchain fundamentals, including consensus mechanisms, cryptography, and distributed ledger technology
- KU2.** Proficiency in smart contract programming and deployment
- KU3.** Experience with development environments like Remix, Truffle, Hardhat, or Brownie
- KU4.** Familiarity with Web3.js, ethers.js, or other blockchain interaction libraries

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- KU5.** Understanding of version control system
- KU6.** Knowledge of secure coding practices and vulnerability assessment
- KU7.** Experience with smart contract auditing and risk assessment
- KU8.** Proficiency in testing frameworks and debugging tools specific to blockchain development
- KU9.** Knowledge of blockchain-specific security measures and best practices
- KU10.** Understanding of compliance requirements and security controls for blockchain solutions

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to tackle complex technical challenges in blockchain systems.
- GS2.** Meticulousness in identifying and addressing potential issues.
- GS3.** Flexibility to keep up with rapidly evolving blockchain technologies.
- GS4.** Skill to explain technical concepts to both technical and non-technical stakeholders.
- GS5.** Ability to work effectively with diverse teams, including developers and business analysts.
- GS6.** Capacity to analyze and evaluate different testing approaches and results.
- GS7.** Persistence in debugging and troubleshooting complex blockchain interactions.
- GS8.** Efficiency in prioritizing and executing various testing tasks.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Incorporate regulatory standards that are applicable for the blockchain solution based on its use case (HIPAA for healthcare industry, etc) in the solution design	3	6	-	2
PC2. Ensure compliance with the government regulations, and plan audits/ reviews to monitor the same	3	5	-	2
PC3. Assess the policies, standards and guidelines of the organization applicable to blockchain implementation	3	5	-	2
PC4. Define enterprise processes to ensure satisfactory performance of the solution	2	4	-	2
PC5. Incorporate on-chain and off-chain governance standards	3	5	-	2
PC6. Define a plan for conducting code audits and security compliance, and define standards for the same	3	5	-	2
PC7. Establish governance models for decentralized systems.	3	5	-	2
PC8. Ensure compliance with relevant regulations (e.g., GDPR, KYC/AML	3	5	-	2
PC9. Implement necessary measures for data privacy and protection.	3	4	-	1
PC10. Conduct research to identify the latest developments in blockchain technologies by working with the R&D team	2	2	-	1
PC11. Identify opportunities for blockchain implementation and its adoption to new use cases	1	2	-	1
PC12. Create whitepapers collating the research data	1	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8737
NOS Name	Establish governance models and best practices
Sector	IT-ITeS
Sub-Sector	
Occupation	Blockchain
NSQF Level	6
Credits	2
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

The assessment will consist of a blend of hands-on practical evaluations, viva-voce, and online proctored scenario-based multiple-choice questions ensuring a thorough evaluation of the individual's proficiency in learning outcomes, practical understanding, and real-world application of concepts

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8732.Define project requirements, conduct feasibility studies, and select appropriate blockchain platform.	30	50	-	20	100	15
SSC/N8733.Design overall system architecture, including data flow, consensus mechanism, and node structure.	30	50	-	20	100	15
SSC/N8734.Develop high-level smart contract specifications and security strategies & deploy the blockchain solution	30	50	-	20	100	15
SSC/N8735.Identify vulnerabilities in various layers of blockchain and develop security features for them.	30	50	-	20	100	15
SSC/N8736.Plan for scalability and integration with existing systems	30	50	-	20	100	15
SSC/N8737.Establish governance models and best practices	30	50	-	20	100	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	200	330	-	120	650	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.